跨DC大模型训练仿真实验报告

1. 实验背景与配置

1.1 模型配置: LLaMA3.1 8B

参考链接: [config.json]

```
{
  "hidden_size": 4096,
  "intermediate_size": 14336,
  "model_type": "llama",
  "num_attention_heads": 32,
  "num_hidden_layers": 32,
  "num_key_value_heads": 8,
  "torch_dtype": "bfloat16",
  "vocab_size": 128258
}
```

1.2 训练超参数 (仿真一个训练Iteration)

- TP = 1
- DP = 4
- PP = 4
- Sequence Length = 8192
- GBS = 128 (微调)
- Token/Iteration = 1M
- Micro Batch Size = 2
- 优化器/加速：没有ZeRO，也没有FlashAttention优化

1.3 硬件参数与集合通信

- **NPU算力**: 312 TFLOPS (A100)
- **NPU显存带宽**: 1.56 TB/s (A100)
- **集合通信算法**: AllReduce - Ring (Chunk=4)

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2. 网络拓扑架构

2.1 DC * 1 (单数据中心)

```
graph TD
    subgraph DC0 [Data Center 0]
        DC0_HBW[HBW Switch]
        NPU0_0[NPU 0]
        NPU0_1[NPU 1]
        NPU0_2[NPU 2]
        NPU0_3[NPU 3]
        DC0_LBW[LBW Switch]

        DC0_HBW ---|600GB/s, 10ns| NPU0_0
```

```
DC0_HBW ---|600GB/s, 10ns| NPU0_1
DC0_HBW ---|600GB/s, 10ns| NPU0_2
DC0_HBW ---|600GB/s, 10ns| NPU0_3

NPU0_0 ---|200Gb/s, 10us| DC0_LBW
NPU0_1 ---|200Gb/s, 10us| DC0_LBW
NPU0_2 ---|200Gb/s, 10us| DC0_LBW
NPU0_3 ---|200Gb/s, 10us| DC0_LBW

DC1_HBW[HBW Switch]
NPU1_0[NPU 4]
NPU1_1[NPU 5]
NPU1_2[NPU 6]
NPU1_3[NPU 7]
DC1_LBW[LBW Switch]

DC1_HBW ---|600GB/s, 10ns| NPU1_0
DC1_HBW ---|600GB/s, 10ns| NPU1_1
DC1_HBW ---|600GB/s, 10ns| NPU1_2
DC1_HBW ---|600GB/s, 10ns| NPU1_3

NPU1_0 ---|200Gb/s, 10us| DC1_LBW
NPU1_1 ---|200Gb/s, 10us| DC1_LBW
NPU1_2 ---|200Gb/s, 10us| DC1_LBW
NPU1_3 ---|200Gb/s, 10us| DC1_LBW

DC2_HBW[HBW Switch]
NPU2_0[NPU 8]
NPU2_1[NPU 9]
NPU2_2[NPU 10]
NPU2_3[NPU 11]
DC2_LBW[LBW Switch]

DC2_HBW ---|600GB/s, 10ns| NPU2_0
DC2_HBW ---|600GB/s, 10ns| NPU2_1
DC2_HBW ---|600GB/s, 10ns| NPU2_2
DC2_HBW ---|600GB/s, 10ns| NPU2_3

NPU2_0 ---|200Gb/s, 10us| DC2_LBW
NPU2_1 ---|200Gb/s, 10us| DC2_LBW
NPU2_2 ---|200Gb/s, 10us| DC2_LBW
NPU2_3 ---|200Gb/s, 10us| DC2_LBW

DC3_HBW[HBW Switch]
NPU3_0[NPU 12]
NPU3_1[NPU 13]
NPU3_2[NPU 14]
NPU3_3[NPU 15]
DC3_LBW[LBW Switch]

DC3_HBW ---|600GB/s, 10ns| NPU3_0
DC3_HBW ---|600GB/s, 10ns| NPU3_1
```

```
        DC3_HBW ---|600GB/s, 10ns| NPU3_2
        DC3_HBW ---|600GB/s, 10ns| NPU3_3

        NPU3_0 ---|200Gb/s, 10us| DC3_LBW
        NPU3_1 ---|200Gb/s, 10us| DC3_LBW
        NPU3_2 ---|200Gb/s, 10us| DC3_LBW
        NPU3_3 ---|200Gb/s, 10us| DC3_LBW
    end

    Spine[Spine Switch]

    DC0_LBW ---|800Gb/s, 10us| Spine
    DC1_LBW ---|800Gb/s, 10us| Spine
    DC2_LBW ---|800Gb/s, 10us| Spine
    DC3_LBW ---|800Gb/s, 10us| Spine
```

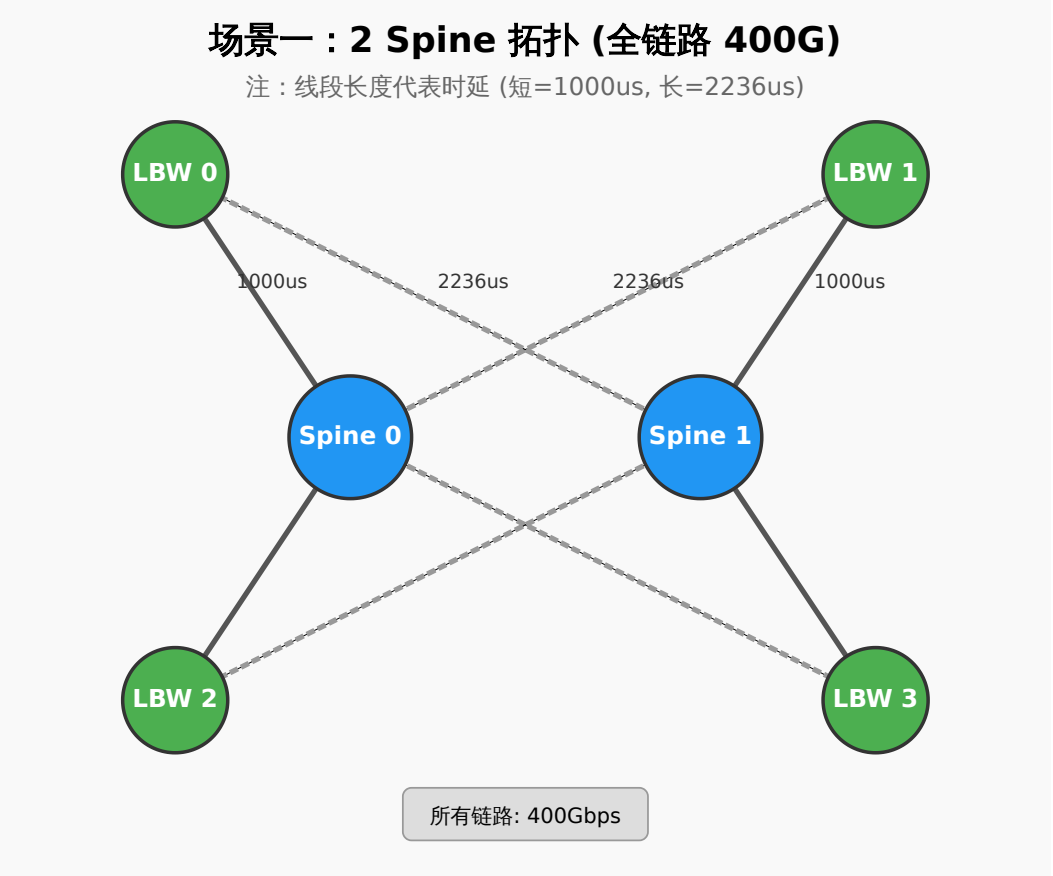
2.2 DC * 4 (跨数据中心)

```
graph TD
    subgraph DCs [Data Centers - Square Layout]
        subgraph TopRow [Top Row]
            subgraph DC0 [Data Center 0]
                DC0_HBW[HBW Switch]
                NPU0_0[NPU 0]
                NPU0_1[NPU 1]
                NPU0_2[NPU 2]
                NPU0_3[NPU 3]
                DC0_LBW[LBW Switch]
                DC0_HBW ---|600GB/s, 10ns| NPU0_0
                DC0_HBW ---|600GB/s, 10ns| NPU0_1
                DC0_HBW ---|600GB/s, 10ns| NPU0_2
                DC0_HBW ---|600GB/s, 10ns| NPU0_3
                NPU0_0 ---|200Gb/s, 10us| DC0_LBW
                NPU0_1 ---|200Gb/s, 10us| DC0_LBW
                NPU0_2 ---|200Gb/s, 10us| DC0_LBW
                NPU0_3 ---|200Gb/s, 10us| DC0_LBW
            end
            subgraph DC1 [Data Center 1]
                DC1_HBW[HBW Switch]
                NPU1_0[NPU 4]
                NPU1_1[NPU 5]
                NPU1_2[NPU 6]
                NPU1_3[NPU 7]
                DC1_LBW[LBW Switch]
                DC1_HBW ---|600GB/s, 10ns| NPU1_0
                DC1_HBW ---|600GB/s, 10ns| NPU1_1
                DC1_HBW ---|600GB/s, 10ns| NPU1_2
                DC1_HBW ---|600GB/s, 10ns| NPU1_3
                NPU1_0 ---|200Gb/s, 10us| DC1_LBW
                NPU1_1 ---|200Gb/s, 10us| DC1_LBW
                NPU1_2 ---|200Gb/s, 10us| DC1_LBW
```

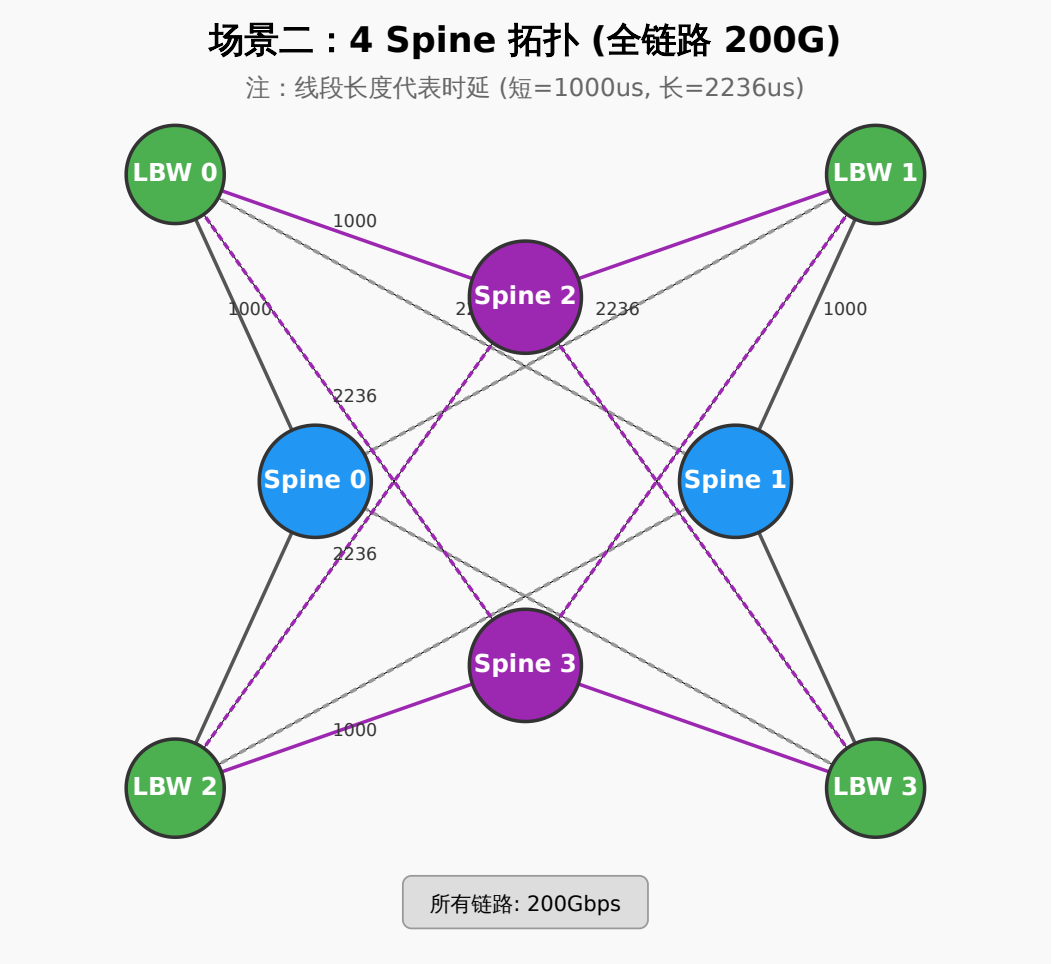
```
            NPU1_3 ---|200Gb/s, 10us| DC1_LBW
        end
    end
    subgraph BottomRow [Bottom Row]
        subgraph DC2 [Data Center 2]
            DC2_HBW[HBW Switch]
            NPU2_0[NPU 8]
            NPU2_1[NPU 9]
            NPU2_2[NPU 10]
            NPU2_3[NPU 11]
            DC2_LBW[LBW Switch]
            DC2_HBW ---|600GB/s, 10ns| NPU2_0
            DC2_HBW ---|600GB/s, 10ns| NPU2_1
            DC2_HBW ---|600GB/s, 10ns| NPU2_2
            DC2_HBW ---|600GB/s, 10ns| NPU2_3
            NPU2_0 ---|200Gb/s, 10us| DC2_LBW
            NPU2_1 ---|200Gb/s, 10us| DC2_LBW
            NPU2_2 ---|200Gb/s, 10us| DC2_LBW
            NPU2_3 ---|200Gb/s, 10us| DC2_LBW
        end
        subgraph DC3 [Data Center 3]
            DC3_HBW[HBW Switch]
            NPU3_0[NPU 12]
            NPU3_1[NPU 13]
            NPU3_2[NPU 14]
            NPU3_3[NPU 15]
            DC3_LBW[LBW Switch]
            DC3_HBW ---|600GB/s, 10ns| NPU3_0
            DC3_HBW ---|600GB/s, 10ns| NPU3_1
            DC3_HBW ---|600GB/s, 10ns| NPU3_2
            DC3_HBW ---|600GB/s, 10ns| NPU3_3
            NPU3_0 ---|200Gb/s, 10us| DC3_LBW
            NPU3_1 ---|200Gb/s, 10us| DC3_LBW
            NPU3_2 ---|200Gb/s, 10us| DC3_LBW
            NPU3_3 ---|200Gb/s, 10us| DC3_LBW
        end
    end
end

Spine[Spine Switch]

DC0_LBW ---|800Gb/s, 1414us| Spine
DC1_LBW ---|800Gb/s, 1414us| Spine
DC2_LBW ---|800Gb/s, 1414us| Spine
DC3_LBW ---|800Gb/s, 1414us| Spine
```

2.3 跨DC多Switch (Spine Switch)

- **2个交换机**


场景一：2 Spine 拓扑 (全链路 400G)
注：线段长度代表时延 (短=1000us, 长=2236us)
LBW 0
LBW 1
1000us
2236us
2236us
1000us
Spine 0
Spine 1
LBW 2
LBW 3
所有链路: 400Gbps

- **4个交换机**



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3. 实验结果与性能分析

本节详细对比了不同的并行策略（数据并行 DP 和流水线并行 PP）在不同网络拓扑下的性能。

3.1 性能数据汇总表

实验场景	并行策略映射	Wall Time (Cycles)	暴露通信时间 (Cycles)	Wall Time相对值	暴露通信时间占比
DC内部 (Intra-DC)	Intra-Node DP + Inter-Node PP	6,119,227,668	242,684,816	1.000x	3.97%
	Intra-Node PP + Inter-Node DP	40,232,773,776	34,356,230,924	6.575x	85.39%
跨DC (Inter-	Intra-Node DP + Inter-	6,152,923,668	276,380,816	1.006x	4.49%

DC)	DC PP				
	Intra-Node PP + Inter-DC DP	77,595,045,399	71,717,126,035	12.681x	92.42%

结果分析：

1. **PP vs DP 对带宽的敏感度**：无论是在单个 DC 内还是跨 DC 场景，将**模型并行 (PP) 放置在节点间/跨DC链路**上，而将**数据并行 (DP) 放置在节点内部**的性能表现远好于相反的策略。这是因为 DP (AllReduce) 会产生突发的巨大流量，更适合高带宽、低延迟的节点内网络；而 PP (P2P通信) 通信量相对较小且易于与计算重叠，适合跨节点/跨DC链路。
2. **跨DC延迟影响**：对比 Intra-DC PP 和 Inter-DC PP，跨 DC 增加的延迟对整体性能 (Wall Time) 影响极小（增加不到1%）。但如果是 Inter-DC DP，则性能下降极其严重（相比 Intra-DC DP 翻倍增加）。

3.2 跨DC多 Spine 交换机 (ECMP) 对比

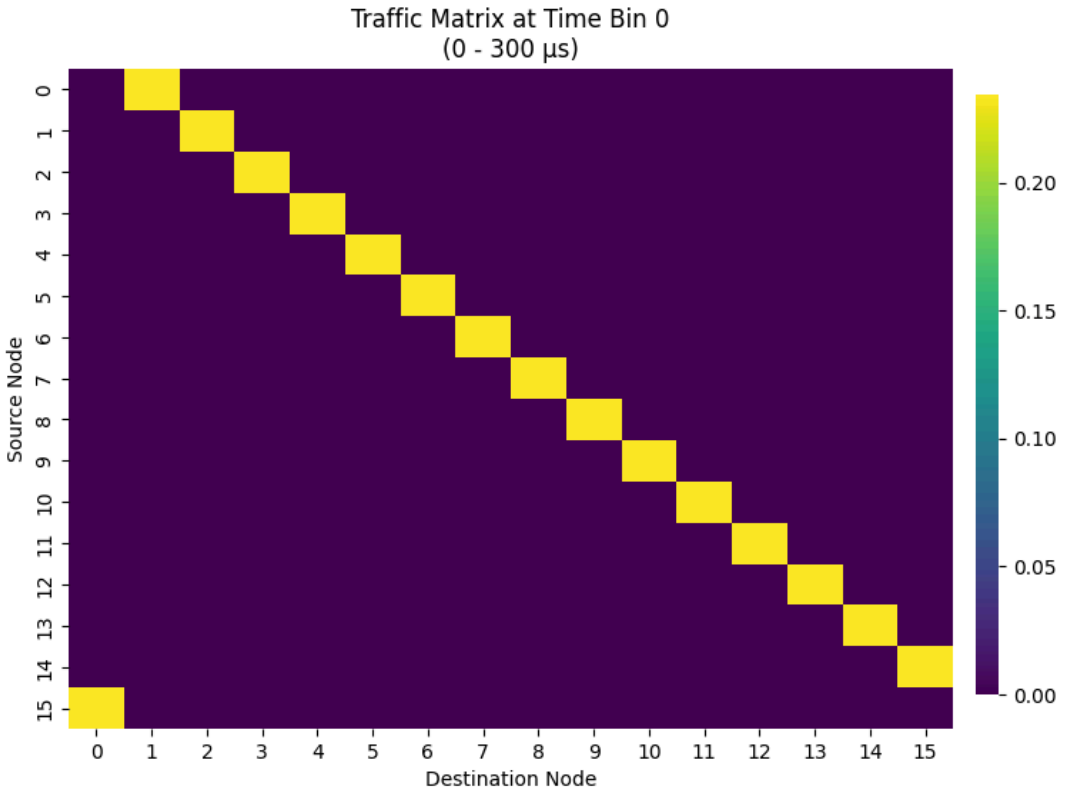
在 **Intra-Node DP + Inter-DC PP** 的最优策略下，改变跨 DC 的 Spine Switch 数量以测试多路径网络对性能的影响：

Spine Switch 数量	Wall Time (Cycles)	暴露通信时间 (Cycles)	Wall Time相对值	暴露通信时间占比
1 个	6,152,923,668	276,380,816	1.000x	4.49%
2 个	6,159,771,808	283,228,956	1.001x	4.60%
4 个	6,189,080,375	312,537,523	1.006x	5.05%

结果分析： 跨DC执行PP，在带宽合理分配的情况下，增加Spine交换机对性能影响不明显。

仿真后验流量矩阵

- 只验证了仿真环境一个环形AllGather算法的流量矩阵，可视化后的流量矩阵



TODO

- [ ] 仿真LocalSGD进行跨DC训练的DP策略，减少DP产生的通信负载，测试性能
- [ ] TE?